# AI Speech and language variation & change

A study on manifestation of AI speech in human natural language

class: [16838] germanische sprachen im vergleich
tutor: Hüning / Konvicna
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## Preliminary

inspired by BSI (2025) and Yakura et al. (2025) which found evidence for GPT influenced human language after the introduction of chatGPT we tried to replicate the Yakura et al. (2025) pipeline of building an AI vocabulary (model preferred lemmata) and compare frequencies of model typical words across pre- and post model introduction human language corpora. The first draft essai proves their hypothesis that LLM generated language manifests within human natural language.

## p2

descriptive stats: GPT vocab raw frequencies

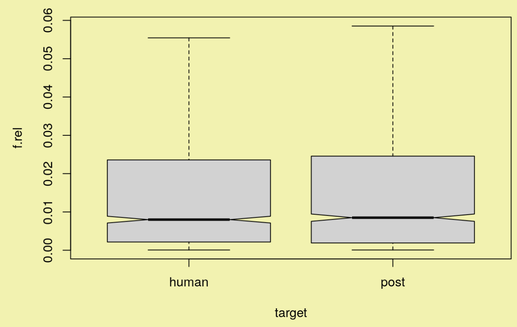


Figure 1: stats descriptive

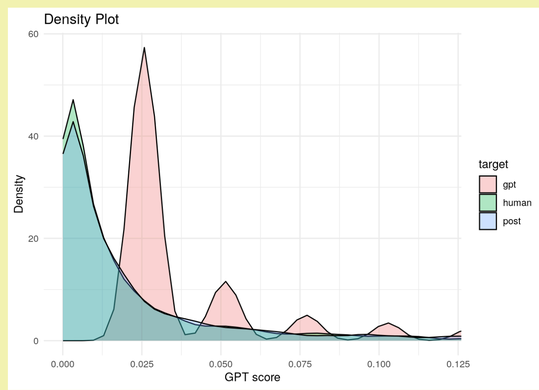


Figure 2: model vocab density

GPT scores vs relative frequency by target

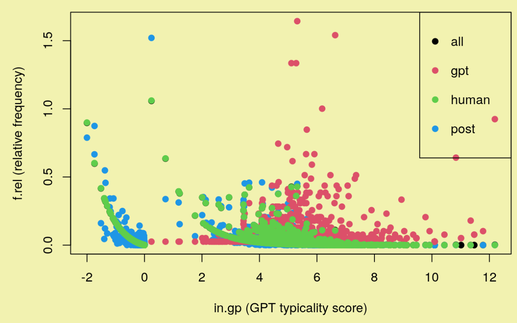


Figure 3: model vocab distribution

## Methods / Data

our human languaga data consists of raw texts from german bundestag plenary protocols, DIP (2026). the LLM corpus consists of model summaries of a first subset of these texts generated with the prompt you see in c1.

we first devised model typical lemmata in the AI corpus which are distinct for that corpus using a linear regression model that calculates a score for each lemma in the corpus, see Figure 3 and Figure 2.

## c1

```
System prompt: You are a member
of german parliament. Prepare a
summary of the text provided to
present at a local community
meeting of your party members.
Output in german language, no
preamble, no extra information,
just the plain text. Wordcount
maximal 300 words, containing
not more than 5% of the keywords
of the text provided and
explicitly not just a list of
keywords but an entertaining
```

## c2

table sdf

| | target | freq |
|---|---|---|
| 1 | human | 0.3708 |
| 2 | post | 0.3890 |
| 3 | DIFF: | 0.0182 |

## Results

to gather an insight, yet with simple descriptive stats comparing the raw frequencies of gpt-preferred lemmas in pre- and post-gemini onset we find that in the target corpus the occurences of these lemma increase, see Figure 1.

## Questions / Hypotheses

- do humans adapt to language produced by LLM and incorporate model vocabulary into their own language production?
- H1: we will find higher frequencies of LLM typical vocabulary within human natural language corpora after onset of model introduction.

### References